

NDM_Setup (Zero-RISC)

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Outlines

- Challenges
- Synthesis
- Inquiries

Challenges

- Errors while reading LEF file.
- Unable to find 2D ports.

Errors while reading LEF file.

Challenges

```
Error: Line 175943, Macro 'SAEDHVT14_TIEDIN_PVIECO_6', Layer 'DIFF_18' does not exist in the library technology information. (LEFR-021)
Information: Changed the site definition for the SAEDHVT14_TIEDIN_PVIECO_6 library cell to unit from none. (LEFR-064)
Warning: Line 175876, 'SYMMETRY' is not defined for marco 'SAEDHVT14_TIEDIN_PVIECO_6' , only 'N' rotation is allowed. (LEFR-037)
Information: Line 175876, Set is_zero_spacing for 9/9 non-real cut OBS in macro SAEDHVT14_TIEDIN_PVIECO_6. (LEFR-062)
Error: Line 176070, Macro 'SAEDHVT14_TIEDIN_VIECO_6', Layer 'DIFF_18' does not exist in the library technology information. (LEFR-021)
Information: Changed the site definition for the SAEDHVT14_TIEDIN_VIECO_6 library cell to unit from none. (LEFR-064)
Warning: Line 176013, 'SYMMETRY' is not defined for marco 'SAEDHVT14_TIEDIN_VIECO_6' , only 'N' rotation is allowed. (LEFR-037)
Information: Line 176013, Set is_zero_spacing for 9/9 non-real cut OBS in macro SAEDHVT14_TIEDIN_VIECO_6. (LEFR-062)
Error: read lef file '/mnt/hgfs/GP/saed14_pdk/stdcell_hvt/lef/saed14nm_hvt_lp9m.lef' failed. (LEFR-002)
design SAEDHVT14_ADDF_V1_0P5 has no physical library.
design SAEDHVT14_ADDF_V1_1 has no physical library.
design SAEDHVT14_ADDF_V1_2 has no physical library.
design SAEDHVT14_ADDF_V2_0P5 has no physical library.
design SAEDHVT14_ADDF_V2_1 has no physical library.
design SAEDHVT14_ADDF_V2_2 has no physical library.
design SAEDHVT14_ADDH_0P5 has no physical library.
```

Unable to find 2D ports.

Challenges

- During setup of PnR flow, while reading top.sdc

```
Warning: No port objects matched 'crash_dump_o[current_pc][31]' (SEL-004)  
Error: Nothing matched for collection (SEL-005)  
Error: Nothing matched for objects (SEL-005)
```

- In “ibex_top”, output port was of type struct {2D - port}

```
output crash_dump_t crash_dump_o,
```

- Solution: creation of ibex_wrapper to flatten the the struct

```
output logic [31:0] current_pc,  
output logic [31:0] next_pc,  
output logic [31:0] last_data_addr,  
output logic [31:0] exception_pc,  
output logic [31:0] exception_addr,
```

HVT_qor

Synthesis

Timing Path Group 'clk_i'

```
-----
Levels of Logic:                5.00
Critical Path Length:           0.18
Critical Path Slack:            2.64
Critical Path Clk Period:       6.25
Total Negative Slack:           0.00
No. of Violating Paths:         0.00
Worst Hold Violation:           -0.10
Total Hold Violation:           -0.12
No. of Hold Violations:         2.00
-----
```

Timing Path Group 'cluster_clock_gating'

```
-----
Levels of Logic:                72.00
Critical Path Length:           6.04
Critical Path Slack:            0.03
Critical Path Clk Period:       6.25
Total Negative Slack:           0.00
No. of Violating Paths:         0.00
Worst Hold Violation:           -3845.62
Total Hold Violation:           -55590.14
No. of Hold Violations:         2069.00
-----
```

Cell Count

```
-----
Hierarchical Cell Count:        2
Hierarchical Port Count:        807
Leaf Cell Count:                13368
Buf/Inv Cell Count:             2028
Buf Cell Count:                 725
Inv Cell Count:                 1308
CT Buf/Inv Cell Count:          0
Combinational Cell Count:       11348
Sequential Cell Count:          2020
Macro Count:                    0
-----
```

Area

```
-----
Combinational Area:             5335.281577
Noncombinational Area:          2211.031222
Buf/Inv Area:                   476.722785
Total Buffer Area:               233.46
Total Inverter Area:            251.48
Macro/Black Box Area:           0.000000
Net Area:                       10221.824505
-----
Cell Area:                      7546.312799
Design Area:                    17768.137304
```

Inquiries

- Is including 2 ref_libs while using the create_lib command necessary for creating scenarios later in the PnR flow?
- Set_driving_cell is not supported. Can we proceed without it?

```
icc2_shell> read_sdc $SDC
```

```
Information: Loading SDC version 2.1 file '/mnt/hgfs/0_GP/Synthesis/5_7/syn_maxperf_new_0/output/ibex_top.sdc(FILE-007)
```

```
Warning: The -library option of set_driving_cell is not supported.(CSTR-022)
```

```
Warning: The -library option of set_driving_cell is not supported.(CSTR-022)
```

```
Warning: The -library option of set_driving_cell is not supported.(CSTR-022)
```

Thank you